star orbits will also be different from $v = r\omega$. **(Option B is incorrect)**

11 **B**

At point P, $F_{net} = 0$

$$\frac{GM_E m}{r^2} = \frac{GM_M m}{(R-r)^2}$$
$$\frac{r}{R-r} = \sqrt{\frac{M_E}{M_M}}$$

Rearranging and making r the subject,

$$r = \frac{R}{\sqrt{\frac{M_E}{M_M}} + 1}$$

12 A